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Double-Step R1rho-Based Lorentzian Fitting (DROF): A New CEST Analysis Approach and Its Comparison With Existing Methods

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ABSTRACT

At 3-T MRI, reliable quantification of CEST contrast is challenged by peak overlap and broad background contamination. Because of this limitation, some previous studies employ four-pool Lorentzian fitting (i.e., amide, direct water saturation, magnetization transfer, and relayed nuclear Overhauser enhancement) for CEST analysis at 3 T, where the multiple CEST contrasts at positive frequency offsets are treated as a single pool. In addition, for fitting-based methods, the fitting accuracy can be readily affected by initial values and boundary conditions. To address these problems, a new CEST analysis approach, named double-step R1rho-based Lorentzian fitting (DROF), with initial value randomization, was developed in this study for extracting five-pool (additional CEST@2 ppm pool) Lorentzian fitting parameters at 3 T. Moreover, a comprehensive evaluation framework including three metrics was proposed to assess the performance of CEST analysis methods. The accuracy and robustness of DROF were evaluated using both simulation and in vivo human data at 3 T, and its performance was compared with existing methods. Results demonstrate that DROF offers improved specificity and stability, suggesting its potential to enhance CEST applicability at 3 T.

1 | Introduction

Chemical exchange saturation transfer (CEST) imaging has emerged as a pivotal molecular imaging modality for in vivo metabolic profiling. By detecting proton exchange between

solute molecules and bulk water, CEST facilitates noninvasive characterization of key biochemical contrasts, including amide proton transfer (APT) at 3.5 ppm [1, 2], which is indicative of cellular protein content and pH. Significant advancements have been made in the application of APT-MRI for tumor grading

Abbreviations: BME, Bloch–McConnell equation; CEST, chemical exchange saturation transfer; CSF, cerebrospinal fluid; DMPLF, double-step multi-pool Lorentzian fitting; DWS, direct water saturation; DROF, double-step R1rho-based Lorentzian fitting; GM, gray matter; LDA, Lorentzian difference analysis; LDA2, LDA with MT pool included in the Z_{ref} fitting; MPLF, multi-pool Lorentzian fitting; MTR_{asym} , magnetization transfer ratio asymmetry; MT, magnetization transfer; PLOF, polynomial and Lorentzian line-shape fitting; rNOE, relayed nuclear Overhauser enhancement; SROF, single-step R1rho-based Lorentzian fitting; WM, white matter; WSP, weak saturation pulse.

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[3, 4], tumor recurrence detection [5, 6], and the assessment of acute ischemic stroke [7, 8]. Moreover, other CEST contrasts have been developed to investigate various types of endogenous molecules. Relayed nuclear Overhauser enhancement (rNOE) can be employed to probe aliphatic protons of mobile lipids and proteins in tumors [9–11]. Glucose CEST (glucoCEST) or dynamic glucose-enhanced (DGE) MRI techniques have the capability to map glucose utilization within the body in vivo [12–14]. CEST at 2 ppm (CEST@2ppm) has been shown to enable quantitative imaging of guanidyl protons in creatine within muscle and adipose tissue [15–17].

Despite notable advancements, challenges remain in CEST quantification. A key issue is peak overlap at clinical magnetic field strengths (e.g., ≤ 3 -T MRI) and interference from broad background signals [18, 19] induced by magnetization transfer (MT) contrast. The conventional magnetization transfer ratio asymmetry (MTR_{asym}) method [20], while effective in eliminating symmetric direct water saturation (DWS) and MT effect [21], still suffers from contamination by asymmetric MT effect and upfield rNOE signals [22]. Model-free methods, such as the three-offset method [23] and double saturation power (DSP) approach [24, 25], reduce MT contributions by constructing specialized reference Z-spectra but at the expense of attenuating the interested CEST signals. These two methods demonstrate high computational efficiency, but their accuracy highly depends on assumptions like clear CEST peak boundaries and fast-exchange approximation.

Alternatively, model-based approaches, derived from the analytical solution of the Bloch–McConnell equation (BME) [26], use Lorentzian functions to quantify the DWS or MT effect to enhance the specificity of CEST quantification [27]. However, the accuracy of these model-based methods is heavily dependent on the chosen fitting model. For instance, Lorentzian difference analysis (LDA) [28, 29], which fits only the DWS effect for background signal, suffers from significant MT contamination. Multi-pool Lorentzian fitting (MPLF) [30–32] struggles to accurately resolve the amide signal at 3.5 ppm from CEST signal at 2 ppm, particularly at 3 T, where both signals are often treated as a single pool [33]. Polynomial and Lorentzian line-shape fitting (PLOF) [34], utilizing the R1rho model, reduces spillover effects from the background. However, in vivo results indicate that the polynomial order affects the fitting accuracy of the background signal around 2 ppm [35]. Moreover, low-order polynomials ($\leq$ fourth order) fail to fit both downfield and upfield background signals simultaneously. Therefore, currently PLOF cannot be used to quantify rNOE.

Another significant challenge in CEST quantification is the lack of standardization in methods for both algorithm implementation and performance evaluation [36, 37]. The absence of a ground truth for CEST information in in vivo data [38] is a major obstacle to the standardization of CEST quantification. While phantom studies can quantitatively assess the consistency of CEST quantification results, it has been reported that compounds such as creatine exhibit substantial differences in exchange rates between in vivo and phantom settings [39, 40]. This discrepancy may limit the direct applicability of phantom-based conclusions to in vivo studies. Although it is feasible to

indirectly assess the accuracy of CEST quantification through disease data and corresponding pathological information, contradictory MPLF results have been reported in studies of diseases like stroke [41–44]. Additionally, advanced fitting methods remain susceptible to the influence of initial fitting values and boundary conditions [45], undermining the reproducibility of experimental results.

In this study, a novel CEST analysis method, dubbed the double-step R1rho-based Lorentzian fitting (DROF) approach, with initial value randomization, was developed for the simultaneous extraction of multiple CEST contrasts, including amide, CEST@2ppm (e.g., creatine), rNOE, and MT at 3 T. The effectiveness of utilizing the double-step fitting and initial value randomization strategies to enhance the reproducibility of CEST analysis was explored. The accuracy and robustness of the DROF method were evaluated using both simulation and in vivo human data at 3 T, and its performance was compared with other existing methods based on a proposed evaluation framework including three evaluation metrics.

2 | Theory

The acquired MRI signal $S_{\text{sat}}(\Delta\omega)$ at frequency offset $\Delta\omega$ is first normalized by the signal without saturation S_0 acquired at large frequency offset (e.g., 200 ppm), as follows:

$$Z(\Delta\omega) = \frac{S_{\text{sat}}(\Delta\omega)}{S_0} \quad (1)$$

where $Z(\Delta\omega)$ represents the Z-spectrum value at $\Delta\omega$, ranging from 0 to 1. The peaks in the spectrum reflect the presence of exchangeable protons, as shown in Figure 1. Typically, the CEST peak in Z-spectra can be calculated by a simple subtraction that can eliminate the symmetric DWS and partial MT effects, known as MTR_{asym} :

$$\text{MTR}_{\text{asym}}(\Delta\omega) = Z(-\Delta\omega) - Z(\Delta\omega) \quad (2)$$

While MTR_{asym} cannot handle the asymmetric MT effect and the presence of upfield CEST effect (e.g., rNOE), the Lorentzian line shape was proved to be effective in separating multiple CEST contrasts from the contamination. In fact, the analytical expression of the Z-spectrum can be derived from the BME under specific approximations, which is a Lorentzian line shape, and the solution can be divided into two types based on the introduced approximation assumptions and the solution approaches: the steady state-based solution and the R1rho-based solution.

2.1 | Lorentzian Line Shape on Steady-State-Based Solution

In a two-site BME model with bulk water and solute pool, when an off-resonant radio frequency (RF) with frequency offset $\Delta\omega$ (relative to water's Larmor frequency) is applied, the normalized reduced water z-magnetization $Z(\Delta\omega)$ induced by the solute pool or bulk water (corresponding to proton

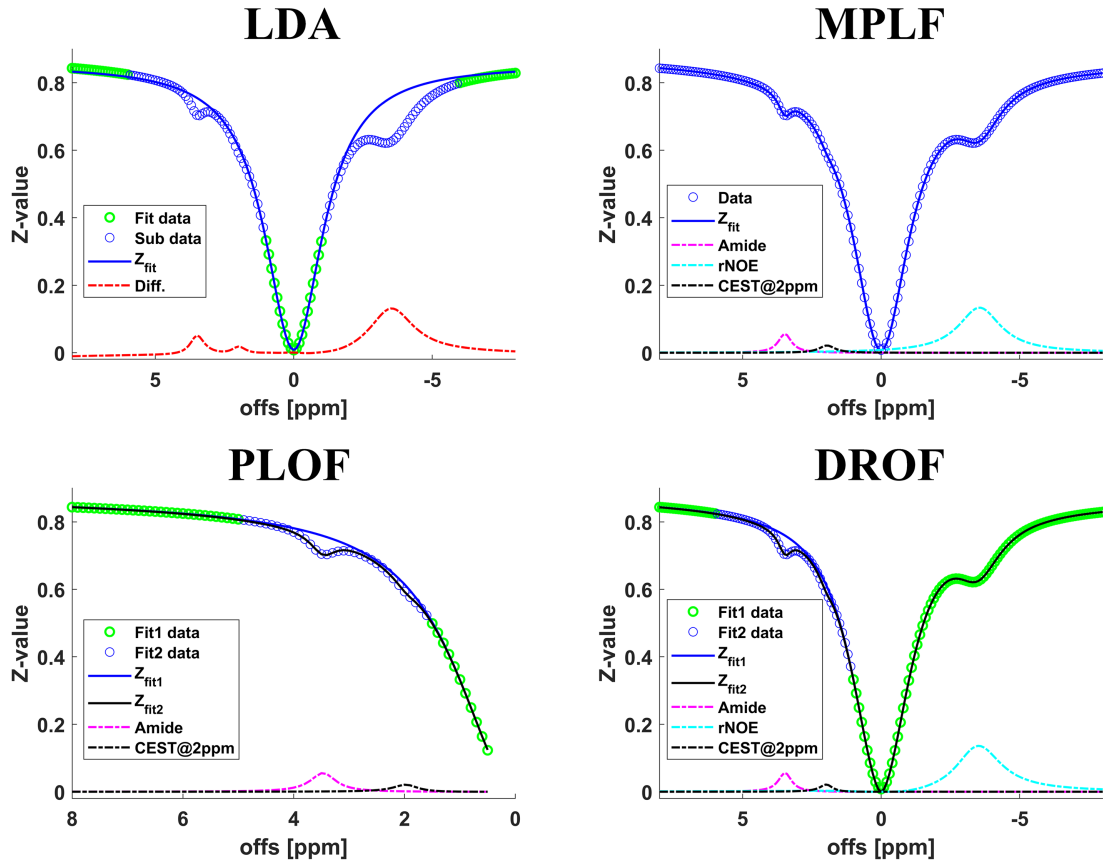


FIGURE 1 | Demonstration of four fitting algorithms, including LDA, MPLF, PLOF, and DROF. Five simulation parameters (T_1 [s], T_2 [s], exchange rate [Hz], $\delta\omega$ [ppm], and fraction [mM]) for each exchange pool are water (1, 0.04, 1, 0, and 111,000), MT (1, 0.00004, 30, -2.5, and 15,000), amide (1, 0.1, 50, 3.5, and 300), CEST@2ppm (1, 0.1, 100, 2, and 100), and rNOE (1.3, 0.005, 20, -3.5, and 2000) [46], respectively. Fit data: data for reference signal fitting in LDA; Sub data: data for CEST contrast subtraction in LDA; and Fit 1/Fit 2 data: data used in the first/second fitting step. DROF, double-step R1rho-based Lorentzian fitting; LDA, Lorentzian difference analysis; MPLF, multi-pool Lorentzian fitting; MT, magnetization transfer; PLOF, polynomial and Lorentzian line-shape fitting; and rNOE, relayed nuclear Overhauser enhancement.

transfer rate (PTR) and DWS, respectively) can be derived from the steady-state condition (i.e., assuming the first-order time-domain derivative of the BME equation to zero) and the weak-saturation-pulse (WSP) assumption (i.e., neglecting the water-pool RF terms or solute-pool RF terms) [47]. Without considering the mutual interference (also known as the spill-over effect) between PTR and DWS, the Z-spectrum under the combined contributions of two pools can be expressed as follows [48]:

$$Z(\Delta\omega) = 1 - \frac{k_{ws}}{R_{1w} + k_{ws}} \cdot \frac{\omega_1^2}{\omega_1^2 + pq + (\Delta\omega - \delta\omega_s)^2 \cdot \frac{q}{p}} - \frac{\omega_1^2}{\omega_1^2 + PQ + (\Delta\omega - \delta\omega_w)^2 \cdot \frac{Q}{P}} \quad (3)$$

with

$$\begin{aligned} p &= R_{2s} + k_{sw} - \frac{k_{sw}k_{ws}}{R_{2w} + k_{ws}}, & q &= R_{1s} + k_{sw} - \frac{k_{sw}k_{ws}}{R_{1w} + k_{ws}} \\ P &= R_{2w} + k_{ws} - \frac{k_{ws}k_{sw}}{R_{2s} + k_{sw}}, & Q &= R_{1w} + k_{ws} - \frac{k_{ws}k_{sw}}{R_{1s} + k_{sw}} \end{aligned} \quad (4)$$

where $R_{1/2,s/w}$ is the relaxation parameter, k_{ws} is the exchange rate from bulk water to solute and vice versa, $\delta\omega_{s/w}$ is the Larmor frequency of exchange pool relative to water, and ω_1 is the Rabi frequency of the RF pulse.

From the Z-spectrum expression obtained through the steady-state approximation, it can be observed that both the bulk water term and solute term can be reformatted into a Lorentzian line-shape form $L(A, \Gamma, \Delta\omega - \delta\omega_p)$, as follows:

$$L(A, \Gamma, \Delta\omega - \delta\omega_p) = \frac{A \cdot \Gamma^2 / 4}{\Gamma^2 / 4 + (\Delta\omega - \delta\omega_p)^2} \quad (5)$$

with peak amplitude A , full width at half maximum (FWHM) Γ , and peak offset $\delta\omega_p$. Therefore, the experimentally acquired Z-spectrum can be quantitatively analyzed using one or multiple Lorentzian functions.

For LDA method, the DWS effect is first fitted to obtain a reference spectra Z_{ref} describing the background. Then, the acquired Z-spectrum $Z(\Delta\omega)$ is subtracted from the fitted reference spectra to calculate CEST contrasts [28]:

$$\text{CEST}(\Delta\omega) = Z_{\text{ref}}(\Delta\omega) - Z(\Delta\omega) = 1 - L_{\text{DS}} - Z(\Delta\omega) \quad (6)$$

An improved LDA approach involves incorporating the MT pool into the Z_{ref} fit, which helps minimize MT interference in the derived CEST contrasts [49].

For MPLF method, the acquired Z-spectrum is quantified as the linear combination of multiple Lorentzian functions [32]:

$$Z(\Delta\omega) = 1 - \sum_{i=1}^n L_i \quad (7)$$

2.2 | Lorentzian Line Shape on R1rho-Based Solution

In addition to deriving the Z-spectrum expression from the steady-state condition, the Z-spectrum can also be expressed using the rotating frame approach. Similar to relaxation under resonance conditions, experiments have shown that the longitudinal magnetization vector under off-resonance conditions can be measured using a similar bi-exponential expression to obtain the characteristic relaxation rate constant $R_{1\rho}$ in the rotating frame, where the effective field is tilted by an angle $\theta = \tan^{-1}(\omega_1 / \Delta\omega)$ off the z-axis. While the magnetization vector precesses around the effective field, the dynamic decay can be analyzed through the average effect of its projection onto the effective field. Assuming the initial magnetization vector is M_0 and the steady-state magnetization is M_{ss} , the averaged Z-magnetization M_z at transient state under off-resonant RF with frequency offset $\Delta\omega$ can be written as follows [50]:

$$M_z(t) = (M_0 \cos^2 \theta - M_{ss})e^{-R_{1\rho}t} + M_{ss} \quad (8)$$

and the steady-state magnetization M_{ss} can be expressed as follows [51]:

$$M_{ss} = \frac{M_0 R_{1w} \cos^2 \theta}{R_{1\rho}} \quad (9)$$

where R_{1w} is the longitudinal relaxation rate of water, while $R_{1\rho}$ is the longitudinal relaxation rate with RF saturation. The Z-spectrum after applying an off-resonant RF pulse with duration t_{sat} is given as follows:

$$Z(\Delta\omega) = \frac{M_z(t_{\text{sat}})}{M_0} = \left(\cos^2 \theta - \frac{R_{1w} \cos^2 \theta}{R_{1\rho}} \right) e^{-R_{1\rho}t_{\text{sat}}} + \frac{R_{1w} \cos^2 \theta}{R_{1\rho}} \quad (10)$$

CEST numerical simulation results indicate that the maximum real eigenvalue (which is negative) of the BME coefficient matrix dominates the exponential decay of the magnetization vector, which is identical to $-R_{1\rho}$ [52]. By simplifying the expression for this special eigenvalue and introducing the asymmetric-population condition between the solute and bulk water, $R_{1\rho}$ can be further expanded as follows [53]:

$$R_{1\rho} \approx R_{1w} \cos^2 \theta + R_{2w} \sin^2 \theta + \frac{f_w f_s (\delta\omega_w - \delta\omega_s)^2 (k_{ws} + k_{sw}) \sin^2 \theta}{\omega_1^2 + (k_{ws} + k_{sw})^2 + (\Delta\omega - \delta\omega_s)^2} \quad (11)$$

$$= R_{1w} + \frac{(R_{2w} - R_{1w}) \omega_1^2}{\omega_1^2 + (\Delta\omega - \delta\omega_w)^2} + \frac{f_w f_s \omega_1^2 (k_{ws} + k_{sw})}{\omega_1^2 + (k_{ws} + k_{sw})^2 + (\Delta\omega - \delta\omega_s)^2} \cdot \frac{(\delta\omega_s - \delta\omega_w)^2}{\omega_1^2 + (\Delta\omega - \delta\omega_w)^2}$$

where f_s/f_w is the population of the exchange pool. In the large-shift assumption where $|\delta\omega_s - \delta\omega_w| \gg \omega_1$, the scaling factor $\frac{(\delta\omega_w - \delta\omega_s)^2}{\omega_1^2 + (\Delta\omega - \delta\omega_w)^2}$ can be regarded as a constant value [51, 54].

Consequently, the second and third terms in the aforementioned expression both exhibit a Lorentzian line-shape form. In this study, the experiments were conducted at a 3-T scanner with a saturation power $B_1 = \omega_1 / \gamma$ less than 1 μT . For amide protons with a Larmor frequency of approximately $3.5 \text{ ppm} \times \gamma B_0$, the large-shift assumption may be partially satisfied with $\frac{\omega_1}{|\delta\omega_s - \delta\omega_w|} = \frac{1}{3.5 \times 3} \ll 1$. Therefore, the amide peak can

be reasonably described using a Lorentzian line shape. It should also be noted that significant deviations from the large-shift assumption may arise when the main field (B_0) is low, the solute pool is close to the water pool (i.e., $\delta\omega_s$ approaches zero), or the saturation power (B_1) is large. These deviations can make Lorentzian line-shape fitting challenging for CEST peak analysis.

For PLOF method, contributions from CEST pools (e.g., amide and creatine) as well as bulk water are quantified as a linear combination of Lorentzian functions while the MT effect is described by an Nth-order polynomial function [35]:

$$R_{1\rho}(\Delta\omega) = C'_0 + L_{\text{water}}(\Delta\omega) + \sum_{n=0}^N C_n \cdot \Delta\omega^n + L_{\text{amide}}(\Delta\omega) + L_{\text{creatine}}(\Delta\omega) \quad (12)$$

where C'_0 and C_n are the coefficients to be fitted.

For our proposed DROF method, we employed the Lorentzian function instead of the polynomial function to quantify MT pool, and the rNOE pool at the upfield is included as an additional Lorentzian term:

$$R_{1\rho} = C'_0 + L_{\text{water}} + L_{\text{rNOE}} + L_{\text{MT}} + L_{\text{amide}} + L_{\text{creatine}} \quad (13)$$

A double-step fitting strategy [55] is employed to improve the fitting accuracy for exchange pools with low concentrations. First, measured Z-spectrum data at ($\Delta\omega \leq 1 \text{ ppm}$ and $\Delta\omega \geq 6 \text{ ppm}$) are used for a three-pool fitting (DWS, MT, and rNOE). Second, a five-pool fitting on all data is performed with fixed first-step-fitted parameters. While the PLOF method utilizes a similar double-step fitting strategy to initially fit the background signal in regions without CEST peaks, the DROF approach expands this initial fitting region to encompass the entire negative frequency offset range. This broader Z-spectrum dataset enables a more robust characterization of the MT line shape and rNOE peak quantification.

3 | Methods

3.1 | CEST Settings

The CEST saturation module employed a continuous wave (CW) pulse with a saturation power (B_1) of 0.8 μT and a saturation time of 2s, as default parameters unless otherwise specified. Two distinct frequency offset schemes were implemented: (1) 161 offsets spaced at 0.1-ppm intervals ranging from -8 to 8 ppm and (2) 51 offsets covering -100 to 100 ppm , with finer sampling

of 0.25-ppm intervals between -4 and 4 ppm. The M_0 signal/image is acquired at -300 ppm additionally.

3.2 | Numerical Simulation

The simulation for 3-T CEST data was performed by numerically solving the BME using the matrix exponential format [56, 57]. Repetition time (TR) was not considered in the simulation, and the magnetization vector was initialized at the beginning of each frequency offset. A random simulated dataset consisting of 5000 Z-spectra was generated using the chemical exchange parameters listed in Table 1. The parameters were selected based on previous studies [39, 58, 59].

3.3 | In Vivo MRI

The in vivo study was approved by the local institutional review board, and the written informed consents were obtained from all participants. Eight asymptomatic volunteers (one female and seven males, aged 74, 73, 71, 71, 71, 74, 76, and 72 years old) were recruited for the in vivo study. The scanning plane was positioned according to the AC-PC (anterior commissure-posterior commissure) line. CEST scans were performed on the GE Signa 3-T scanner using 2D single-shot fast spin echo (SSFSE), with the following parameters: TR = 3 s, echo time (TE) = 20.5 ms, flip angle (FA) = 90° , field of view (FOV) = 220×220 mm², and nominal resolution = $1.72 \times 1.72 \times 6$ mm³. T_1 mapping was performed using dual flip angle (DFA) [60] method with the following parameters: FOV = 220×220 mm², nominal resolution = $1.72 \times 1.72 \times 6$ mm³, TR/TE = 30/1.82 ms, and two FAs of 7° and 20° . T_1 -weighted 3D MPRAGE images were acquired for brain segmentation with the following parameters: FOV = $256 \times 256 \times 256$ mm³, nominal resolution = $1 \times 1 \times 1$ mm³, and TR/TE = 4.86/1.70 ms.

3.4 | Data Analysis

Data analysis was conducted using custom MATLAB code (MathWorks, United States). Eight CEST analysis methods were used: MTR_{asym} , LDA, LDA with MT pool included in the Z_{ref} fitting (LDA2), MPLF, MPLF with double-step fitting strategy (DMPLF), DROF, single-step R1rho-based Lorentzian fitting (SROF), and PLOF using first-order polynomial [61]. The double-step fitting strategy for DMPLF, PLOF, and DROF was

realized by fitting the amide peak and CEST@2 ppm peak using the Z-value from 1 to 6 ppm during the second fitting stage, while fitting the background or other pools using the remaining Z-value during the first fitting stage. For LDA, the DWS pool was fitted using the Z-value at $|\Delta\omega| < 1$ and $|\Delta\omega| > 10$ ppm [28]. For LDA2, the DWS and MT pools were fitted using the Z-value at $|\Delta\omega| < 1$ and $|\Delta\omega| > 10$ ppm. Random initial guesses were realized by the MultiStart function [62] with five guesses unless specified otherwise. Specifically, MultiStart generates a set of random initial values that satisfy the defined fitting boundary conditions, followed by the fitting process for each initial guess [63]. The fitting result with the smallest residual is then selected. The fitting boundaries and initial values are listed in Table S1. The T_1 -weighted MPRAGE images were first co-registered to the CEST M_0 images, followed by brain segmentation using SPM12 (statistical parametric mapping, <https://www.fil.ion.ucl.ac.uk/spm/>) to obtain masks of cerebrospinal fluid (CSF), gray matter (GM), and white matter (WM).

3.5 | Evaluation Metrics

Three evaluation metrics based on the simulation dataset were utilized for CEST analysis methods' quantitative comparison: correlation factor R^2 , variance contribution, and sensitivity. In simulation experiments, the correlation factor was calculated by multiple linear regression between CEST analysis fitting outputs and predefined concentrations of exchange pools. The variance contribution of each calculated concentration variable was calculated using analysis of variance (ANOVA), reflecting the independence of analysis results. The sensitivity was determined by calculating the linear slope between observed CEST pool signals and predefined concentrations. Because the CEST peak amplitude outputs of the R1rho-based algorithm are in the unit of ms^{-1} , their fitted results were first converted to Z-spectra, and the amplitude of a CEST peak was calculated by subtracting the fitted Z-spectra from the fitted background Z-spectra; finally, the linear slope between the amplitude of the CEST peak and the corresponding predefined concentrations was derived.

3.6 | Experiment

3.6.1 | Experiment I. Impact of Double-Step Fitting and Initial Value Randomization Strategies

The impact of double-step fitting and initial value randomization strategies on amide and CEST@2 ppm analysis results was examined through simulation using 161 frequency offsets. The results of the best and worst initial values were selected from the 50 random initial guesses. The scatter plot and the correlation factor R^2 were used to visualize the influence of fitting strategies. The variance contribution was further analyzed to investigate the improvement brought by the double-step fitting strategy.

3.6.2 | Experiment II. Performance on Simulated Data

The performance of five CEST analysis methods (MTR_{asym} , LDA, MPLF, DMPLF, and DROF) was compared for amide, CEST@2 ppm, and rNOE quantification through scatter plot

TABLE 1 | Exchange parameters setting of random simulated dataset using the five-pool Bloch-McConnell equation.

Pools	T_1 (s)	T_2 (s)	k_s (s ⁻¹)	$\delta\omega_p$ (ppm)	f (mM)
Water	1	0.04	1	0	111,000
MT	1	0.00004	20–60	−2.5	5000–25,000
Amide	1	0.1	20–50	3.5	20–400
Creatine	1	0.1	200–500	2	10–100
rNOE	1.3	0.001	10–30	−3.5	100–2000

visualization and variation contribution analysis, based on simulations employing 51 frequency offsets. Furthermore, method robustness was assessed across 49 discrete B_1 values (from 0.1 to 2 μ T) and 48 Gaussian noise levels σ (from 0 to 0.003). Finally, quantitative validation was performed by comparing algorithm-derived apparent exchange-dependent relaxation (AREX) values with corresponding theoretically calculated $R_{\text{ex}}^{\text{CEST}}$ values [49]:

$$R_{\text{ex}}^{\text{CEST}} = f_s k_{\text{sw}} \frac{(\gamma B_1)^2}{(\gamma B_1)^2 + (R_{2s} + k_{\text{sw}}) k_{\text{sw}}} \quad (14)$$

$$\text{AREX}(\Delta\omega) = \left(\frac{1}{Z(\Delta\omega)} - \frac{1}{Z_{\text{ref}}} \right) \cdot (R_{1w} + f_{\text{MT}} R_{1,\text{MT}})$$

For AREX value of amide@3.5 ppm ($\Delta\omega = 3.5$ ppm), the reference signals Z_{ref} were method dependent: (1) MTR_{asym} used Z (−3.5 ppm) as reference; (2) LDA employed the background signal value at 3.5 ppm obtained from Lorentzian fitting; and (3) MPLF, DMPLF, and DROF utilized the fitted Z -spectrum value at 3.5 ppm after excluding the amide pool contribution. The default exchange parameters for the $R_{\text{ex}}^{\text{CEST}}$ simulation are listed in Table S2.

3.6.3 | Experiment III. Comparison on In Vivo Data

The in vivo measurements were conducted using the 51 offsets same as Experiment II. For B_0 correction, the B_0 shift of each voxel was first calculated by minimum search on the cubic spline-interpolated Z -spectra, and then the measured Z -spectra were shifted so that the Z -value at 0 ppm is corresponding to the minimum point. The amide, CEST@2 ppm, and rNOE maps were calculated from five CEST analysis methods (MTR_{asym} , LDA, MPLF, DMPLF, and DROF). The MT maps from MPLF and DROF (not available in MTR_{asym} and LDA) were also generated for comparison. For each algorithm's results, statistical analysis of CEST contrast across different brain regions was performed using two-way ANOVA in GraphPad Prism software (La Jolla, California, United States) to evaluate compound distribution characteristics.

3.6.4 | Experiment IV. Digital Brain Phantom Experiment

Because in vivo human brain CEST data lacks ground-truth exchange parameters (such as concentration and exchange rate) for each exchange pool, a digital brain simulation was further designed to mimic Z -spectrum patterns similar to in vivo brain data, with known ground-truth exchange parameters. First, a single pixel was manually selected within each tissue mask (CSF, GM, and WM), and the Z -spectra of its 24 neighboring pixels were averaged to obtain the representative Z -spectra for that tissue. Next, tissue-specific exchange parameters were carefully optimized to accurately reproduce the in vivo Z -spectra characteristics of each tissue type, with all applied parameters listed in Table S3. Finally, Gaussian noise ($\sigma = 0.002$) was introduced to the simulated data to mimic experimental conditions.

4 | Results

4.1 | Experiment I. Impact of Double-Step Fitting and Initial Value Randomization Strategies

The fitting scatter plot and the R^2 of fitting results using different fitting strategies are shown in Figure 2. For MPLF results, significant R^2 variance is shown between the best and worst initial values (from 0.772 to 0.015 for amide and from 0.728 to 0.088 for CEST@2 ppm). The initial-value sensitivity is also prominent for the PLOF algorithm (from 0.606 to 0.045 for amide and from 0.874 to 0.004 for CEST@2 ppm). For all five algorithms, the results from the best initial value exhibit high linearity in regression, underscoring the importance of initial value selection. However, the optimal initial value is unknown in actual fitting. Therefore, the initial value randomization strategy can be employed to increase the chances of selecting the optimal initial value and reduce the probability of selecting the worst initial value, thereby enhancing the reproducibility of the CEST analysis.

The performance comparison between single-step and double-step fitting strategies can also be found in Figure 2. Specifically, the discrepancies between MPLF and DMPLF, as well as SROF and DROF, demonstrate enhanced linearity in the regression when employing the double-step fitting approach. Because the MT and DWS effect occupy a great portion in the Z -spectra and usually serve as a background signal, the accuracy of the MT and DWS quantification will greatly influence the extraction of other CEST pools with less than 1% relative concentration. Consequently, the double-step fitting strategy that fits the background signal in the first step can not only improve the best initial value results for amide and CEST@2 ppm but also greatly enhance the performance of the worst initial value results (e.g., for amide, from 0.015 in MPLF to 0.776 in DMPLF).

The barplots of variance contribution on amide and CEST@2 ppm results are shown in Figure 3. The MTR_{asym} shows obvious rNOE contamination in both amide and CEST@2 ppm results (60.7%–66.1% variance contribution from rNOE).

For the LDA algorithm, prominent MT contamination exists with a 61.7%–80.0% variance contribution. The LDA2 method incorporating Lorentzian line shape for the MT pool significantly reduces the MT contribution to less than 1.7%; however, the rNOE pool contributes over 22.0% to amide and 51.9% to CEST@2 ppm. The MPLF results show 77.2% amide specificity and 72.7% CEST@2 ppm specificity, while the DMPLF using a double-step fitting strategy increases the CEST@2 ppm specificity to 88.0%. PLOF results show 78.6% amide specificity and 85.7% CEST@2 ppm specificity, while the DROF method reaches the highest specificity with 79.7% in amide and 91.1% in CEST@2 ppm.

The improvement from MPLF to DMPLF, as well as from SROF to DROF, confirms the effectiveness of the double-step fitting strategy in reducing the impact of MT and rNOE pools. Figure 3 demonstrates that the three R1rho-based methods (PLOF, SROF, and DROF) exhibit comparable sensitivity to other approaches.

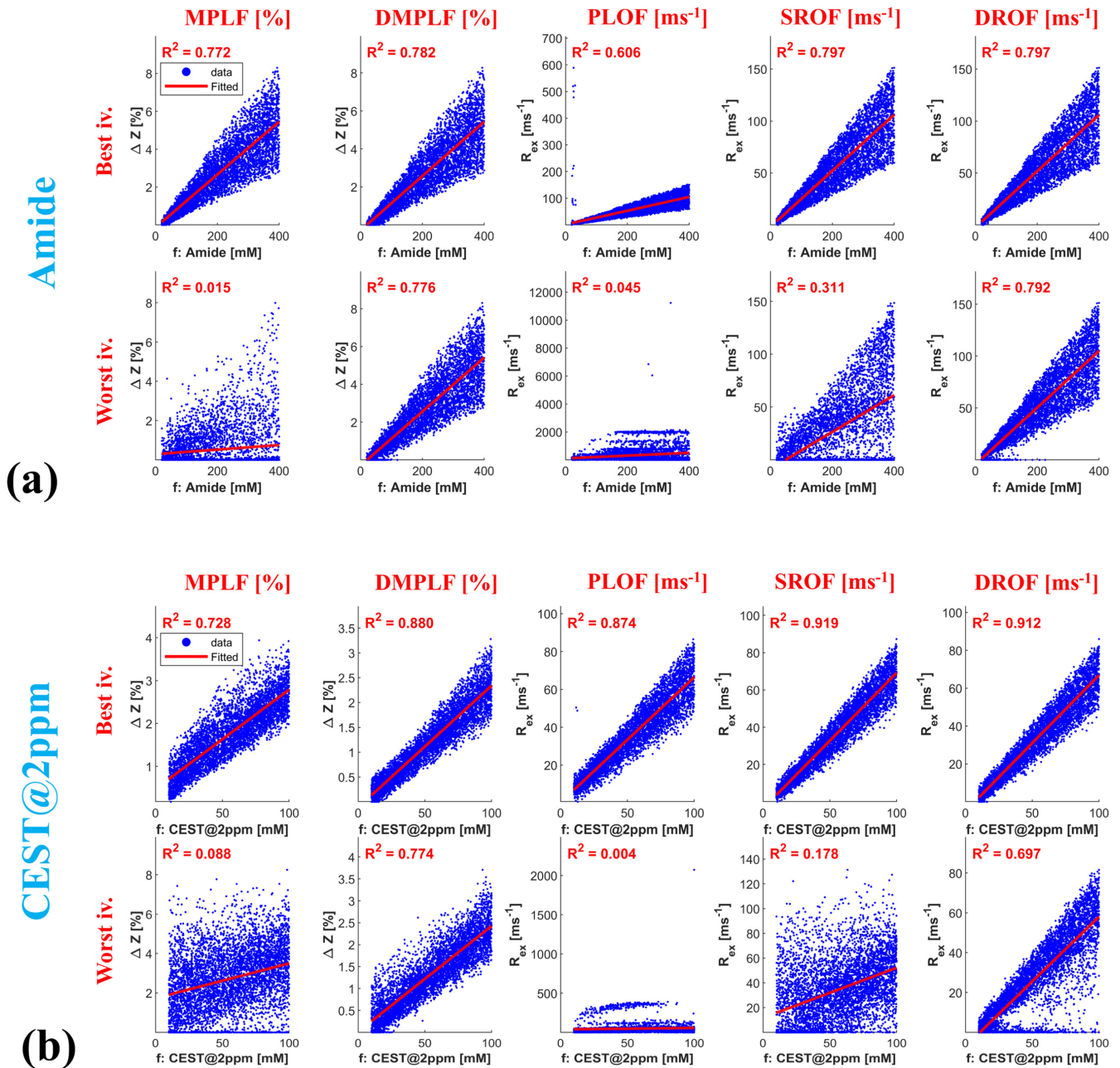


FIGURE 2 | Scatter plot and R^2 of two fitting strategies (random initial guess and double-step fitting) on the fitting results of (a) amide and (b) CEST@2ppm. The “Best iv.” and “Worst iv.” results from the first and the second rows are saved from 50 random initial guesses. The output of MPLF and DMPLF has the same unit of Z-spectra and is thus tagged as “ ΔZ ” with unit percentage. The output of PLOF, SROF, and DROF has the same unit of relaxation and is thus tagged as “ R_{ex} ” with unit ms^{-1} . The dataset is simulated using 161 offsets. iv., initial value; DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; MPLF, multi-pool Lorentzian fitting; PLOF, polynomial and Lorentzian line-shape fitting; and SROF, single-step R1rho-based Lorentzian fitting.

4.2 | Experiment II. Performance on Simulated Data

Figure 4a shows the scatter plots of five CEST analysis methods that are capable of providing amide, CEST@2ppm, and rNOE quantification. When reducing the frequency offset number from 161 in Experiment I to 51 in Experiment II, slight degradation occurs, as shown in Figures 3 and 4b. The rNOE and MT contamination on MTR_{asym} and LDA results can be clearly observed from the scatter plots and variance

contribution bar, while the proposed DROF method achieves the highest specificity for all three CEST contrasts (rNOE, amide, and CEST@2ppm).

Figure 5 demonstrates the performance of five analytical methods under different B_1 values and Gaussian noise levels. First, the four Lorentzian-based methods all exhibit significant variations in R^2 when B_1 ranges from 1.2 to 1.5 μT , reflecting breakdown of the Lorentzian approximation when B_1 exceeds the WSP regime and violates large-shift assumptions.

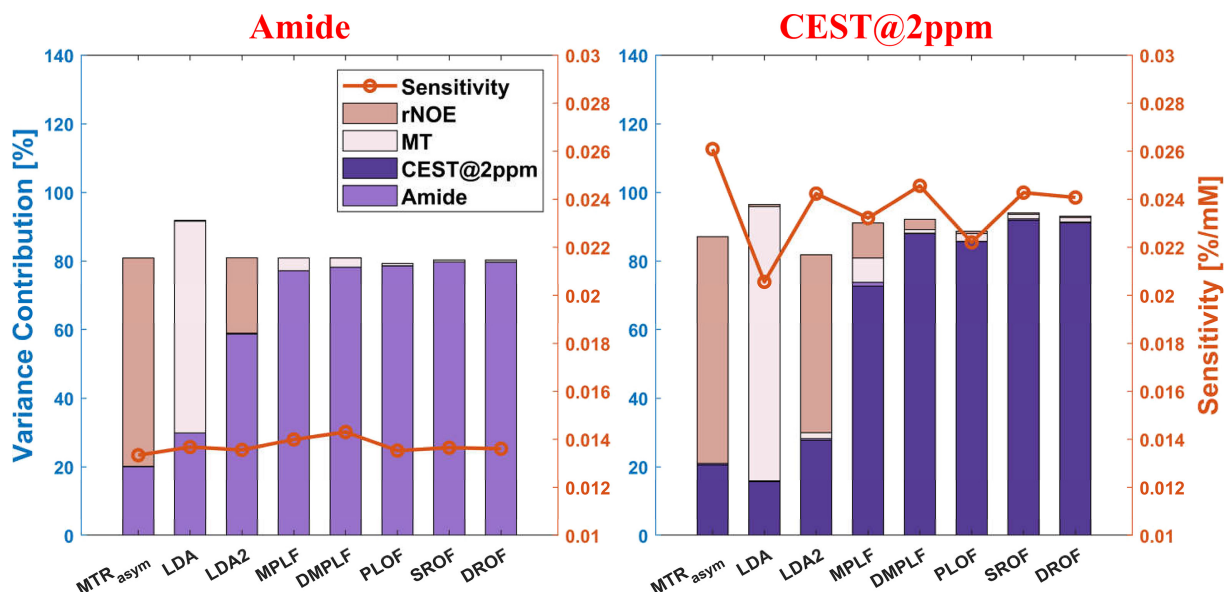


FIGURE 3 | Variance contribution and sensitivity of amide and CEST@2ppm. Total variance contribution less than 100% represents that there are factors (such as the exchange rate of CEST@2 ppm) other than exchange pool fraction that affect the CEST analysis results. The dataset is simulated using 161 offsets. DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; LDA, Lorentzian difference analysis; LDA2, LDA with MT pool included in the Z_{ref} fitting; MPLF, multi-pool Lorentzian fitting; MTR_{asym}, magnetization transfer ratio asymmetry; MT, magnetization transfer; PLOF, polynomial and Lorentzian line-shape fitting; rNOE, relayed nuclear Overhauser enhancement; and SROF, single-step R1rho-based Lorentzian fitting.

Second, within the 0.5–1.2 μT B_1 range, DROF demonstrates superior specificity for all three CEST contrasts, followed by DMPLF and MPLF. In contrast, both MTR_{asym} and LDA show lower specificity because of their limited ability to effectively account for background signals from MT and DWS. Finally, for amide and CEST@2 ppm compounds, DROF's R^2 drops to 0.65 at a Gaussian noise level of $\sigma = 0.002$, performing worse than MPLF, which suggests that DROF is more sensitive to noise.

Figure S1 presents a performance comparison based on exchange-dependent relaxation (R_{ex}) quantification. While $AREX_{LDA}$ demonstrates systematic overestimation and $AREX_{MTR_{asym}}$ tends to underestimate the values, $AREX_{MPLF}$ shows excellent agreement with the theoretically calculated R_{ex} , consistent with prior published results [49]. $AREX_{DROF}$ yields values closely aligned with $AREX_{MPLF}$, further supporting its reliability.

4.3 | Experiment III. Comparison on In Vivo Data

Figure 6 demonstrates CEST contrast maps of Subject 1 calculated by MTR_{asym}, LDA, MPLF, DMPLF, and DROF. Figure 7 presents the results of statistical significance analysis of CEST contrasts across different brain regions in eight subjects. The rNOE maps from all analysis methods consistently show high intensity in WM and low intensity in GM. The MT maps from MPLF and DROF also exhibit similar distributions with higher intensity in WM than in GM. Because of the subtraction principle, the amide map and rNOE map of MTR_{asym} have opposite values, implying the rNOE contamination for MTR_{asym} results of amide. The amide maps from five-pool MPLF and LDA exhibit low intensity in GM, while the amide map from DMPLF

and DROF shows comparable values within WM and GM. For the CEST@2 ppm maps, five analysis methods show substantial differences: MTR_{asym} shows low intensity in WM and high intensity in partial GM region, LDA shows similar distribution with MT maps (calculated by MPLF and DROF), MPLF exhibits low intensity in WM and high intensity in GM, while DMPLF and DROF show no significant difference between GM and WM. Figures S2–S3 display representative contrast maps from two additional subjects, demonstrating consistent effects across individuals.

Figure S4 displays the DFA-calculated R_1 maps and DROF-derived R1rho maps from eight subjects. The quantitative maps demonstrate consistent values across all participants, indicating high measurement reliability. Notably, the R1rho values at -300 ppm closely approximate the R_1 values, as the CEST effect becomes negligible at this offset frequency and the signal is predominantly governed by water proton relaxation. In contrast, significantly elevated R1rho values are observed at -3.5 , 2 , and 3.5 ppm compared to the -300 ppm reference, clearly reflecting the presence of CEST effects. It should be emphasized that R1rho measurements are collectively determined by multiple factors including exchangeable proton concentration, exchange rate, B_1 value, and frequency offset.

4.4 | Experiment IV. Digital Brain Phantom Experiment

Figure 8a,b shows the Z-spectra at three representative voxels for CSF, GM, and WM. The in vivo Z-spectrum of GM exhibits a prominent peak at 2.5 ppm, which may be contributed by phosphocreatine (PCr) or amine. The Z-spectrum in CSF showcases

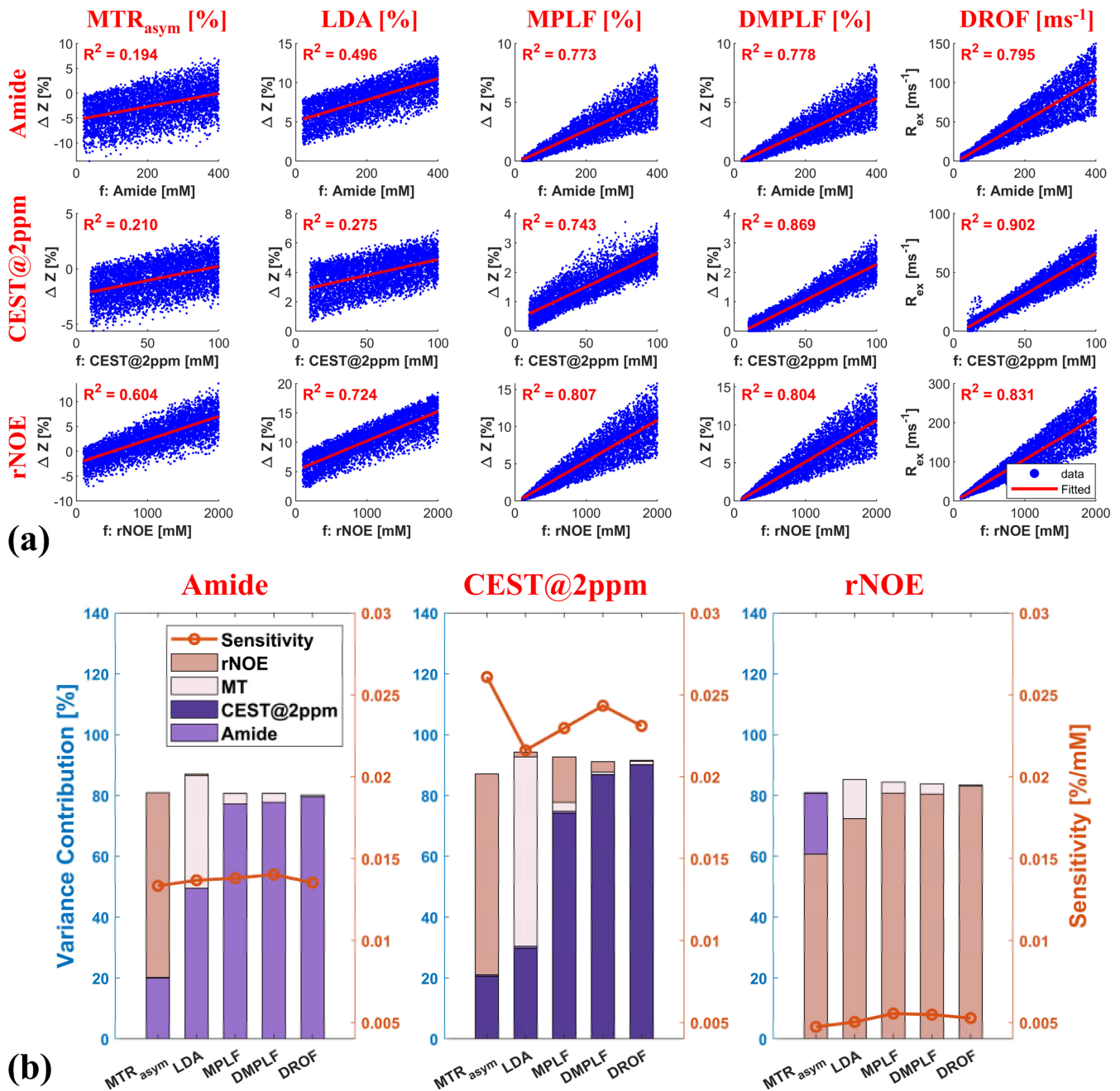


FIGURE 4 | Performance of MTR_{asym}, LDA, MPLF, and DROF upon the analysis of amide, rNOE, and CEST@2ppm. The dataset is simulated using 51 offsets. DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; LDA, Lorentzian difference analysis; MPLF, multi-pool Lorentzian fitting; MTR_{asym}, magnetization transfer ratio asymmetry; MT, magnetization transfer; and rNOE, relayed nuclear Overhauser enhancement.

prominent fluctuation probably due to the flow effect, implying its unreliability for CEST analysis. In Figure 8b, the simulated Z-spectrum manually adjusted using parameters from Table S3 shows reasonable agreement with the measured averaged Z-spectrum, although noticeable discrepancies remain in certain frequency regions. For Z-spectrum in GM and WM, MPLF and DROF both fit well. The MPLF-fitted amide pool Lorentzian coefficients (amplitude, FWHM, and offset of peak in ppm) for in vivo data are (0.019, 0.87, and 3.76) in GM and (0.066, 6.64, and 3.17) in WM. The MPLF-fitted amide pool Lorentzian coefficients for simulation data are (0.0314, 1.012, and 3.55) for GM

and (0.031, 2.96, and 3.72) for WM. The measured Z-spectra and fitted coefficients both indicate a broader bandwidth of amide peak in WM than in GM.

The CEST contrast maps calculated by MTR_{asym}, LDA, MPLF, DMPLF, and DROF using the digital brain phantom with $\sigma=0.002$ Gaussian noise are displayed in Figures 9 and S5. Three key observations emerge from the analysis: (1) Noise robustness evaluation revealed that the MPLF-amide map exhibited significant sensitivity in quantifying amide and CEST@2ppm signals, whereas DROF-calculated maps showed

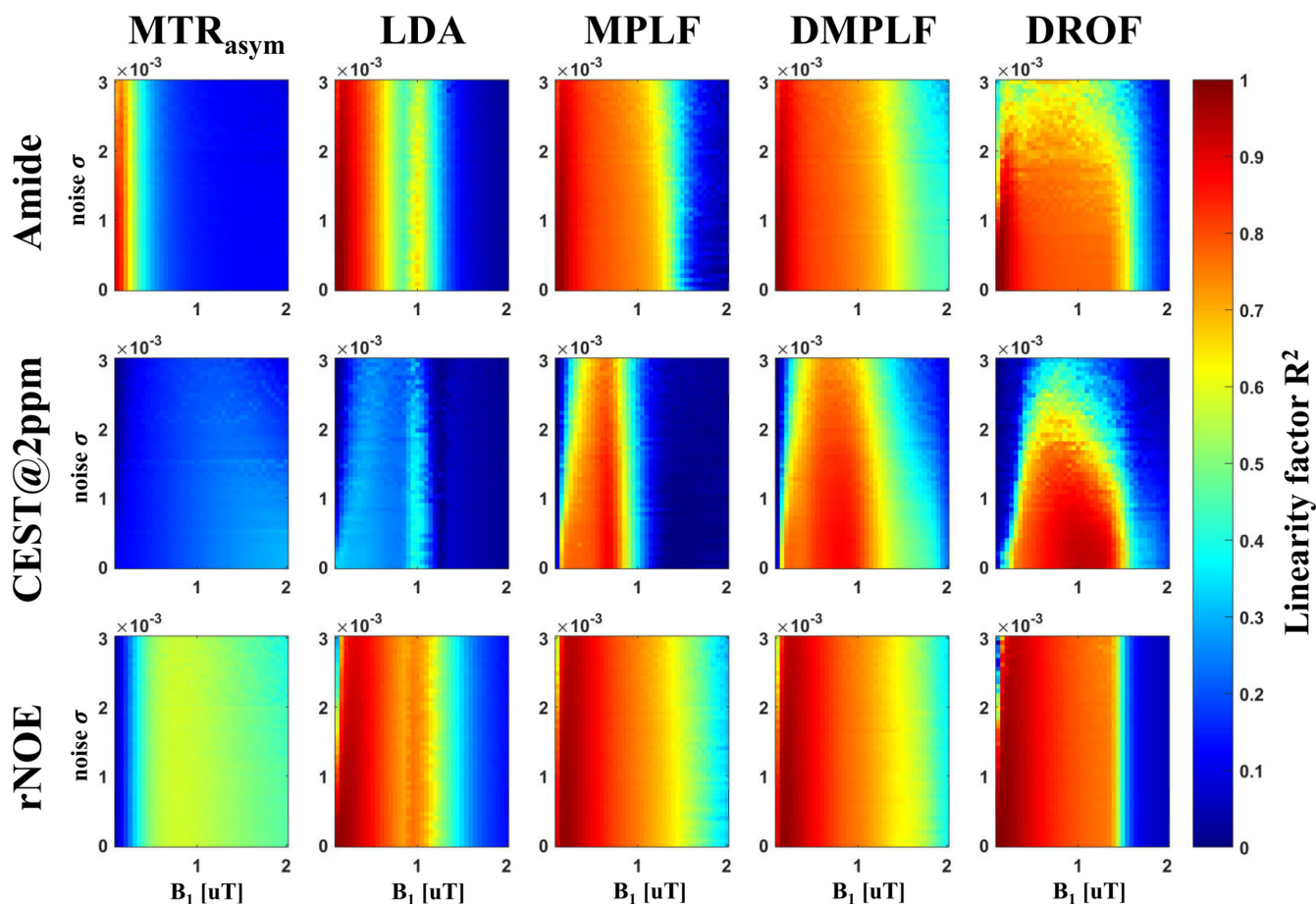


FIGURE 5 | Performance of MTR_{asym} , LDA, MPLF, and DROF upon the analysis of amide, rNOE, and CEST@2ppm, with different saturation power (B_1) and Gaussian noise level. The dataset is simulated using 51 offsets. DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; LDA, Lorentzian difference analysis; MPLF, multi-pool Lorentzian fitting; MTR_{asym} , magnetization transfer ratio asymmetry; MT, magnetization transfer; and rNOE, relayed nuclear Overhauser enhancement.

superior intratissue homogeneity despite residual noise-induced variability in CSF measurements; (2) for amide proton simulation (GM: 110 mM; WM: 350 mM), MTR_{asym} overestimated GM values relative to WM, LDA, and DROF correctly reflected the preset concentration gradient (GM < WM), and MPLF produced comparable intertissue values; and (3) in CEST@2ppm simulation (GM: 50-mM creatine @1.9 ppm + 45-mM phosphocreatine @2.5 ppm; WM: 75-mM creatine @1.9 ppm + 5-mM phosphocreatine @2.5 ppm), MTR_{asym} , MPLF, and DROF consistently yielded higher GM values as expected, while LDA displayed an inverse contrast pattern.

5 | Discussion

This study developed a new CEST analysis method, DROF, for robustly quantifying amide, CEST@2ppm, rNOE, and MT at 3T. DROF is a double-step R1rho-based Lorentzian fitting method that incorporates the random initial value setting. The effectiveness of the initial value randomization and double-step fitting strategy and the superiority of DROF over several widely used line-shape analysis algorithms were verified by both simulation and in vivo experiments. Furthermore, a new evaluation framework including three metrics was proposed in this study

to effectively compare different CEST analysis methods with high reproducibility and intuitive visualization, contributing to the standardization of quantitative CEST.

5.1 | The Impacts of Initial Value Randomization and Double-Step Fitting Strategy

Line-shape fitting methods such as MPLF and PLOF rely on the least square fitting and therefore suffer from the local minimum problem, which makes the results highly sensitive to the selected initial fitting values. This susceptibility reduces the reproducibility of CEST quantification across different scanners and sequence protocols. As shown in Figure 2, there is a substantial reduction in accuracy from the best to the worst fitting results, with this discrepancy being particularly pronounced in the MPLF and PLOF methods. This further highlights the susceptibility of these methods to the selection of initial fitting values, as well as the effectiveness of the initial value randomization in improving fitting accuracy. In addition to the initial fitting values, the fitting boundaries also significantly affect the results of fitting-based methods. Although initial value randomization cannot determine the optimal fitting boundaries, it can effectively mitigate the

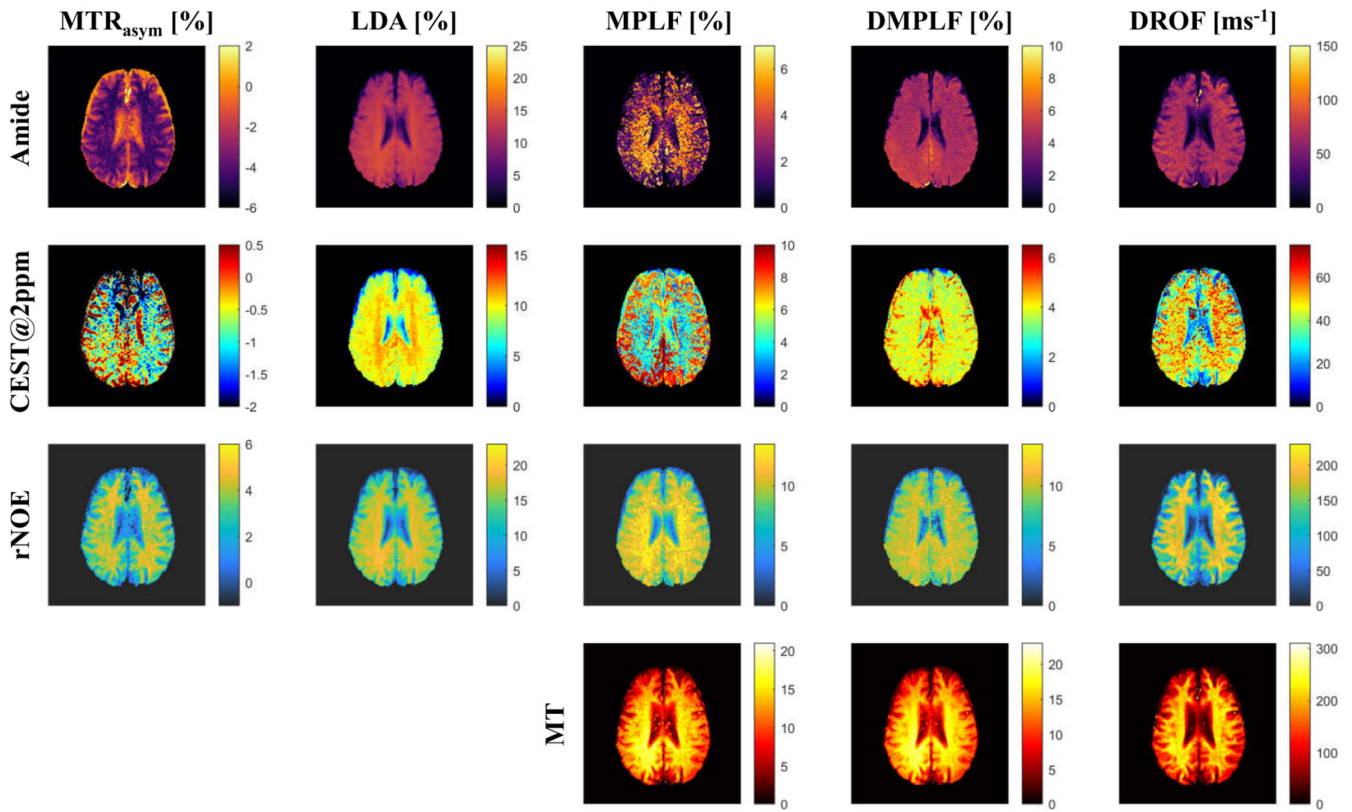


FIGURE 6 | Calculated amide, CEST@2ppm, and rNOE maps of a 74-M asymptomatic volunteer. The CEST sequence used single-shot fast spin echo (SSFSE) readout, with the following parameters: repetition time (TR)=3 s, echo time (TE)=21 ms, field of view (FOV)=220×220×6 mm², matrix size = 256×256×1, and slice thickness=6 mm. M_0 image was acquired at a frequency offset of −300 ppm. DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; LDA, Lorentzian difference analysis; MPLF, multi-pool Lorentzian fitting; MTR_{asym} , magnetization transfer ratio asymmetry; MT, magnetization transfer; and rNOE, relayed nuclear Overhauser enhancement.

influence of local optima. Consequently, the fitting process benefits from wider boundaries without increasing the risk of convergence to suboptimal solutions. Therefore, the influence of boundary selection is indirectly diminished.

It is important to mention that the experiment conducted involved randomly varying the initial values 50 times across a wide fitting range to examine the variation between the best and worst results. This does not imply that using a fixed initial fitting value would always result in such low accuracy. For practical CEST experiments, carefully selected initial values may still yield acceptable fitting results. Furthermore, the initial value randomization requires more computational time for postprocessing and cannot guarantee convergence to the global optimal solution within limited trials. While this strategy helps to statistically reduce the impact of local minimum problems, it cannot guarantee a globally optimal result in all cases. Future studies could explore the feasibility of combining iterative methods [45] or other approaches to further improve the stability of the fitting results.

The comparison between MPLF and DMPLF, as well as between SROF and DROF in Figure 2, demonstrates that the double-step fitting strategy can effectively mitigate the impact of poor initial value setting on the results [64]. In in vivo CEST data, the contributions of DWS and MT effects are predominant in the Z-spectrum, and accurate quantification of these components

greatly influences the extraction of other small CEST peaks. This also explains why PLOF, by using polynomial methods to better remove the MT effect, can more effectively separate CEST@2ppm from other CEST peaks. Because both MT and rNOE peaks are located in the upfield region, some previous studies have used the peripheral region of Z-spectrum data, such as those from ± 14 to ± 100 ppm (which exclude rNOE peaks), to fit the broad MT component for subsequent fitting of CEST peaks [64, 65]. In contrast, the double-step strategy used in this study first fits DWS, MT, and rNOE simultaneously, which allows for more efficient use of the acquired data to fit the background signal. This modified strategy eliminates the need to collect additional peripheral signals while ensuring the accuracy of the MT fitting.

5.2 | CEST Methods' Comparison

For the MTR_{asym} method, which relies on the assumption of asymmetric CEST contrast, the presence of upfield rNOE and MT significantly affects the MTR_{asym} result for downfield CEST peaks at 3.5 and 2 ppm. For the LDA method, which only fits the DWS peak, the asymmetric MT effect strongly influences the CEST peak, as shown in Figure 3, where MT pool accounts for 60% variation contribution of the amide/CEST@2 ppm quantification. In the LDA2 method with the inclusion of MT in the Z_{ref} fitting, the MT contamination is reduced. However, because of

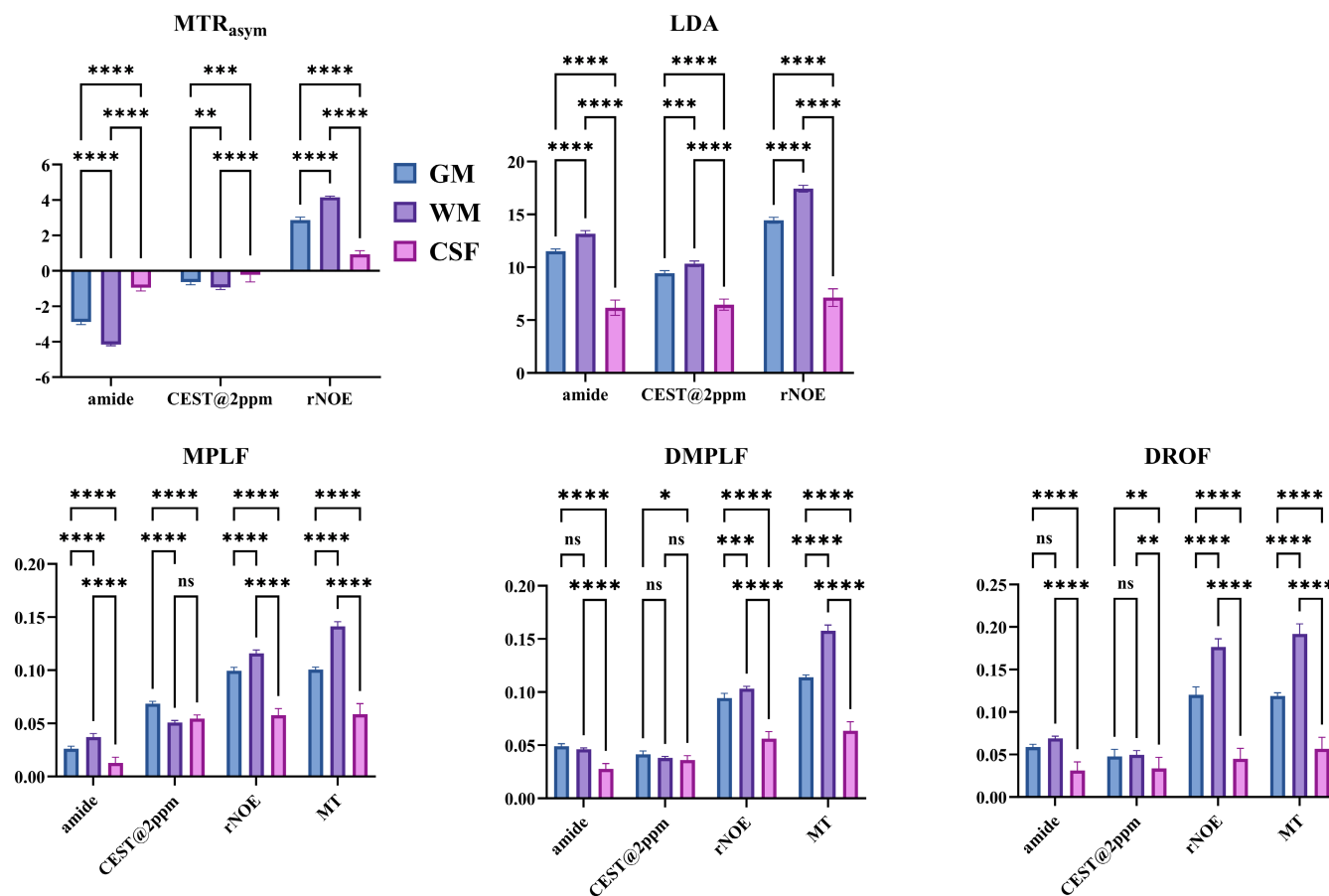


FIGURE 7 | Statistical comparison (eight subjects) of CEST contrasts (amide, CEST@2ppm, rNOE, and MT) using different analysis methods (MTR_{asym} , LDA, MPLF, DMPLF, and DROF) among three brain tissues (GM, WM, and CSF). Significance levels: * $p \geq 0.05$ and $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. A two-way ANOVA was used to compare results from the same analysis method across different brain tissues. CSF, cerebrospinal fluid; DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; GM, gray matter; MPLF, multi-pool Lorentzian fitting; MTR_{asym} , magnetization transfer ratio asymmetry; MT, magnetization transfer; ns, not significant; rNOE, relayed nuclear Overhauser enhancement; WM, white matter.

the overlap between MT at around -2 ppm and rNOE at around -3.5 ppm, the fitting of MT is influenced by rNOE, resulting in substantial rNOE contribution in the LDA2 results. While MTR_{asym} and LDA utilize less CEST data and are computationally simpler, their reliance on prior assumptions necessitates additional caution when interpreting CEST data with MT and rNOE effects.

Figures 3 and 4 show that MPLF outperforms both LDA and MTR_{asym} in terms of fitting quality and specificity on the simulation dataset. Given that the randomized simulation incorporates parameter variations in both concentration and exchange rate (Table 1), the observed $<85\%$ contribution sum for CEST@2ppm in MPLF (Figure 4b) suggests that exchange rate variations may account for over 15% of the quantification error. While Figure 8e,f shows good agreement between MPLF fittings and in vivo Z-spectra, the CEST maps in Figures 6, S2, and S3 display a complementary pattern: reduced amide signals versus elevated CEST@2ppm signals in cortical GM regions. Additionally, the digital brain results in Figure 9 highlight more pronounced noise in the amide map produced by MPLF compared to other methods. These observations suggest that the five-pool MPLF at 3 T may be highly sensitive to the peak overlap of amide and CEST@2ppm. It is

important to note that the larger the exchange rate of a pool, the broader its peak width. While the exchange rate of amide is around 50 s^{-1} , compounds for CEST@2ppm (e.g., creatine) usually have exchange rates in the hundreds, resulting in very broad peaks that significantly overlap with the amide peak. This may explain why most current studies at 3 T using MPLF analyses chose a four-pool model without the consideration of CEST@2ppm [33, 38].

While the large-shift assumption in the R1rho model (PLOF and DROF) is similar to the WSP approximation in the steady-state model (LDA and MPLF), where the interactions between the solute pool and bulk water are both neglected, the R1rho model applies the approximation condition when simplifying the R_{ex} term, whereas the steady-state model applies this approximation condition at the beginning of the derivation. Therefore, the R1rho model tends to have lower accuracy loss and higher robustness against spillover effects [34]. As shown in Figure 2, the DROF method exhibits minimal sensitivity to initial fitting values, while maintaining comparable sensitivity and achieving the highest specificity among all methods (Figures 3 and 4). The stability inherent in the R1rho model allows DROF to extend the quantification of five-pool models to more pools at high fields, with higher accuracy and reproducibility.

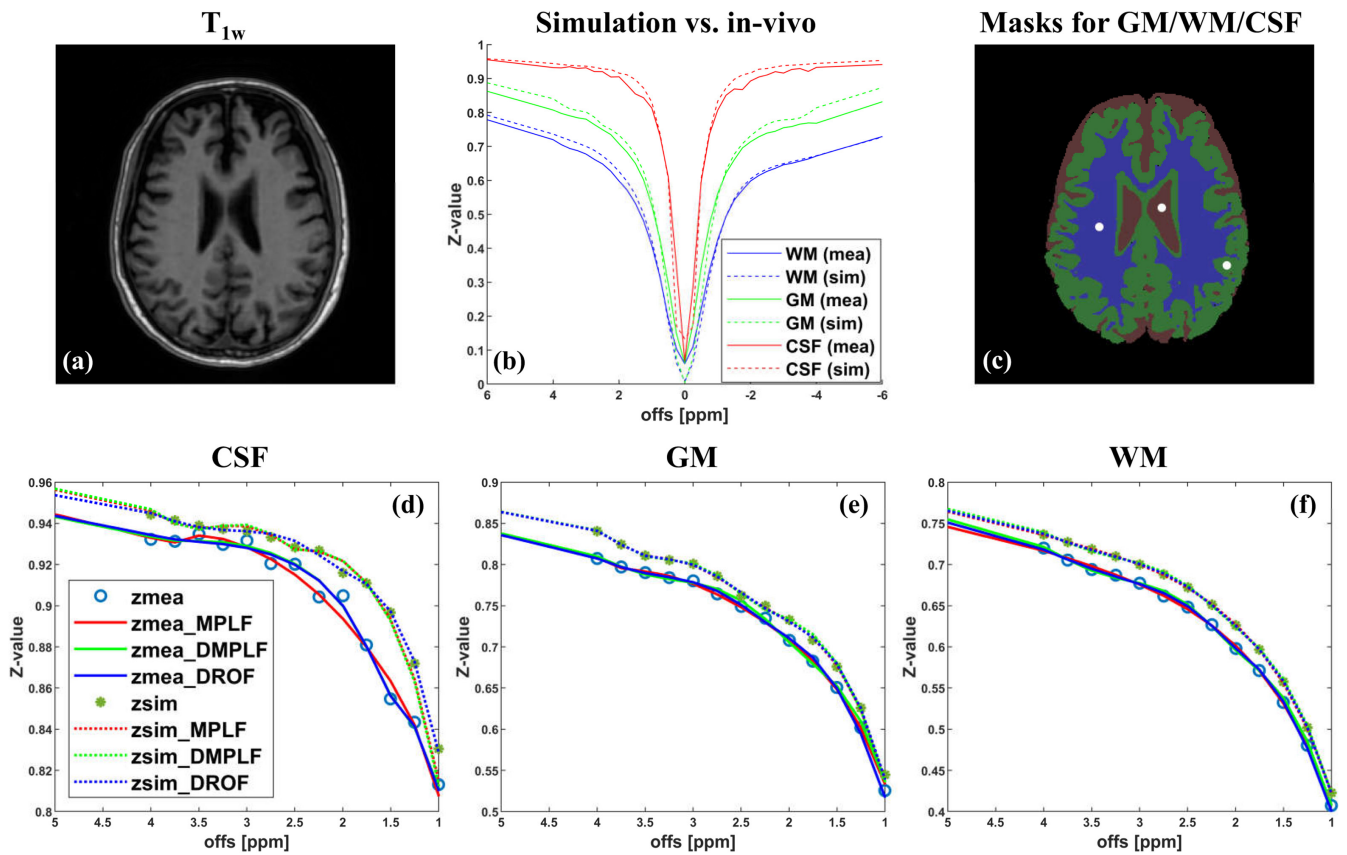


FIGURE 8 | (a) T_1 -weighted image used for brain segmentation; (b) Z-spectra of GM, WM, and CSF regions obtained by simulation (using parameters in Table S3) and in vivo measurement (averaging ROIs in the corresponding brain areas); (c) masks of GM (green), WM (blue), and CSF (red) derived from SPM12 segmentation, with the three white circles indicating manually selected ROIs; (d–f) partial Z-spectra of three regions and the corresponding fitted results from MPLF and DROF. CSF, cerebrospinal fluid; DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; GM, gray matter; MPLF, multi-pool Lorentzian fitting; MT, magnetization transfer; WM, white matter; zmea, measured in vivo Z-spectra; and zsim, simulated Z-spectra.

5.3 | Metrics and Clinical Interpretation

This study takes advantage of simulation datasets with known ground truth, using three performance metrics—correlation factor R^2 , variance contribution, and sensitivity—to quantitatively evaluate CEST algorithms. This evaluation method clearly illustrates the strengths and weaknesses of several commonly used algorithms and also highlights the effectiveness of DROF in improving the stability of fitting results. This simulation-based evaluation method can also be used to quantitatively compare the accuracy of model-free methods (e.g., DSP approach [24]), as well as the effectiveness of data preprocessing methods like quasi-steady state (QUASS) [66] in improving CEST contrast. However, it is important to note that these performance metrics may change depending on the parameters used in the simulation dataset. Future studies could attempt to generate more realistic simulated datasets by fitting in vivo measurements [67], thereby establishing improved ground truth for evaluating CEST algorithm performance.

Figure 7 demonstrates that DROF-derived amide values show no significant differences between WM and GM, consistent with previous reports [59, 68]. However, for CEST@2 ppm quantification, DROF results reveal no WM–GM contrast, which diverges from prior 3-T GuanCEST studies reporting higher values in GM

[69]. It is also worth noting that, while this study mainly focuses on the impact of concentration changes in the exchange pools on calculated peak amplitude, exchange rate could vary in lesions such as tumors and ischemia and greatly change the linewidth of the CEST peaks. Previous studies have shown that different postprocessing algorithms exhibit varying sensitivities to exchange rate changes [25]. In future work, disease data including CEST and other supporting sequences such as magnetic resonance spectroscopy (MRS) [70, 71] can be acquired to further validate the performance of these CEST analysis algorithms.

5.4 | Limitation and Future Directions

Despite its advantages in CEST contrast separation, the DROF method has several limitations that should be addressed in future work. First, previous studies [48, 72] have suggested that more accurate and detailed equations could better describe the semisolid MT pool and its coupling with other pools. Incorporating such models may further reduce the MT contamination on CEST quantification. Second, similar to the PLOF method, DROF employs a complex R1rho model, which results in a fitting time much longer than that of the steady state-based methods. If the number of initial value randomizations increases or if whole-brain data are processed, the computational

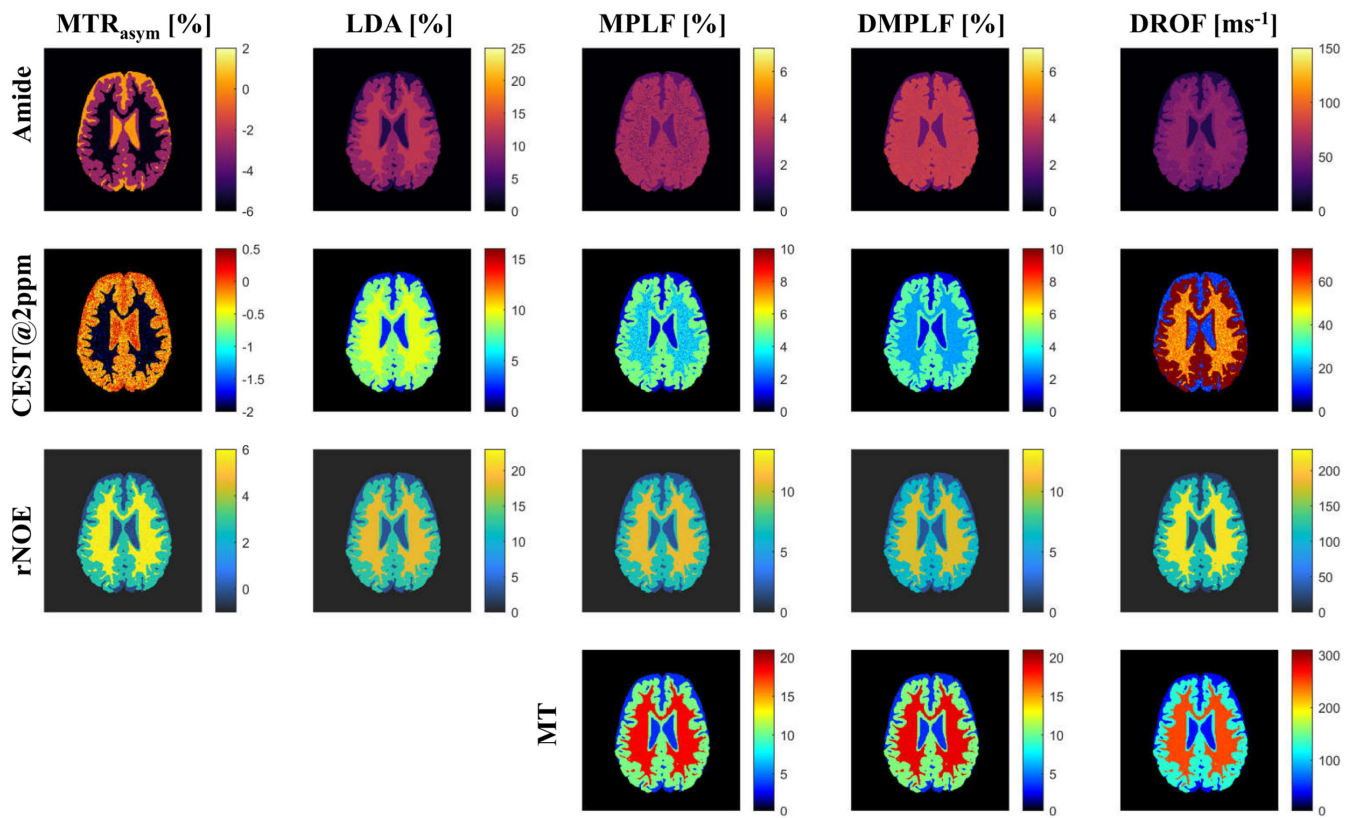


FIGURE 9 | Calculated amide, CEST@2ppm, and rNOE maps from brain phantom with Gaussian noise ($\sigma = 0.002$). DMPLF, double-step multi-pool Lorentzian fitting; DROF, double-step R1rho-based Lorentzian fitting; LDA, Lorentzian difference analysis; MPLF, multi-pool Lorentzian fitting; MTR_{asym} , magnetization transfer ratio asymmetry; MT, magnetization transfer; and rNOE, relayed nuclear Overhauser enhancement.

time will grow even further. Future work could explore parallel computing or deep learning approaches [73] to accelerate the postprocessing. Lastly, the R1rho model in Equation (10) used in DROF may be affected by non-steady-state conditions. The derivation of Equation (8) is based on the spin-lock R1rho theory [50], which requires steady-state acquisition. If the recovery time or the saturation time is not sufficient to reach steady state, deviation may arise, such as the oscillations induced by R2rho. A potential solution is eliminating the impact of insufficient saturation and recovery time using the QUASS method.

The simulation results in Figure 5 demonstrate that DROF exhibits greater sensitivity to Gaussian noise compared to other methods, showing lower R^2 values than MPLF at $\sigma=0.002$. Consequently, we specifically compared the performance at this noise level in the digital brain (Figures 9 and S5), where MPLF unexpectedly displayed higher variation in WM regions. This apparent discrepancy may be attributed to two factors: (1) DROF's increased sensitivity in R^2 calculation (Figure 5) likely stems from the greater complexity of its R1rho model and (2) R^2 , as a statistical metric derived from multiple Z-spectra with randomized exchange parameters, may not directly reflect algorithm performance under fixed-parameter Z-spectrum conditions. In this study, we employed additive Gaussian noise in our experiments. While previous research has demonstrated that MRI image noise can be approximated as Gaussian distribution when signal-to-noise ratio (SNR) is larger than two [74], the actual noise distribution more closely resembles Rician

noise [75]. Future investigations could evaluate DROF's performance by incorporating various CEST-dedicated denoising algorithms [76, 77] under different noise distributions.

The proposed DROF method demonstrates substantial potential for broad applications. Although the current 3-T implementation only employs a five-pool model, the framework can be readily extended to high-field (e.g., 7 T) CEST studies by incorporating additional pools (e.g., amine@3ppm [25] and hydroxyl@1ppm [14, 78]). Crucially, as quantitative CEST (qCEST) techniques like Lorentzian-corrected apparent exchange-dependent relaxation (LAREX) [79] and quantification of exchange rate using varying saturation power (QUESP) [80] fundamentally probe R_{ex} , Figures S1 and S4 demonstrate DROF's capability to accurately derive R_{ex} via R1rho modeling, offering a novel tool for qCEST analysis. Furthermore, DROF's applicability extends beyond neuroimaging to metabolic studies in diverse tissues including intervertebral discs [81], prostate [82], and muscle [17].

6 | Conclusion

This study developed a DROF approach with the initial value randomization strategy for CEST data analysis. Additionally, a new evaluation framework was proposed to compare the performance of different CEST analysis approaches. The effectiveness and stability of the DROF method in simultaneously quantifying amide, CEST@2ppm, rNOE, and MT at 3 T were confirmed

using both simulation and in vivo CEST data of the human brain. Moreover, DROF outperformed other existing CEST analysis methods. The strategies of double-step fitting and initial value randomization have been demonstrated to significantly improve the accuracy and stability of CEST analysis methods. This work highlights DROF as a reliable analysis tool for providing stable and comprehensive CEST analysis results, especially at 3 T.

Acknowledgments

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Data Availability Statement

The CEST analysis algorithm code and the reproduction code are publicly available at <https://github.com/AIMLaboratory/DROF>.

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